

RAMCO INSTITUTE OF TECHNOLOGY
Department of Electronics and Communication Engineering
Academic Year: 2018 - 2019 (Odd Semester)

Innovative Practices Description

Degree, Semester& Branch: VII Semester B.E. ECE A

Course Code & Title: EC6702 Optical Communication and Networks

Name of the Faculty member: Mr.V.Rajesh

Name of the Topic: Modulation of a LED

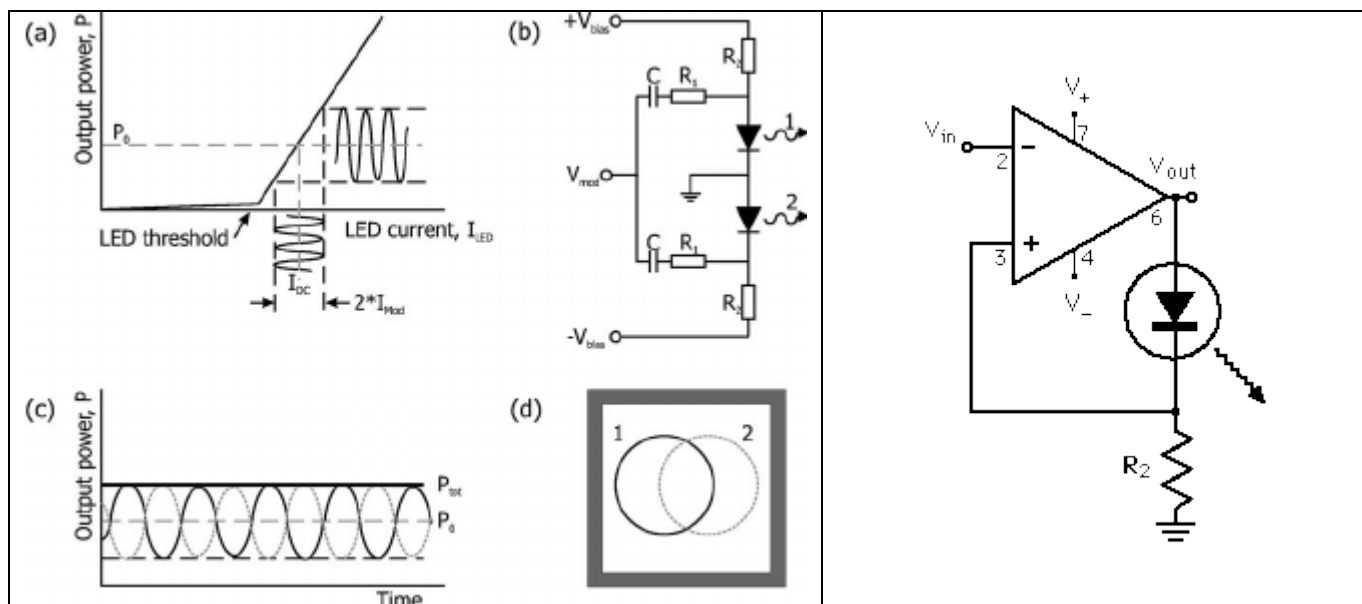
Name of the Innovative Practice: Animated Video

Description:

LED Modulation. Since the light output from an LED has a region where it is linearly proportional to the forward current through the diode, it is useful for producing a light level which is proportional to some signal. That signal can then be send through a fiber optic cable and detected on the other end.

Since the light output from an LED has a region where it is linearly proportional to the forward current through the diode, it is useful for producing a light level which is proportional to some signal. That signal can then be send through a fiber optic cable and detected on the other end. The basic need for modulation is then a device which produces a current proportional to the input voltage of the applied signal. This can be done with an op-amp arranged as a voltage-to-current converter.

LED Modulation Response. ... The LED 3-dB modulation optical bandwidth is defined as the modulation frequency at the LED power transfer function is reduced by 3 dB [1]. The LED 3-dB modulation optical bandwidth could be expressed as: $f_{3dB} = \sqrt{3} / 2\pi / (T_n + TRC_n)$.



Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO3	1	1	2	-	1	1	-	-	-	1	-	1

CO3: The students will be able to Explain the Various optical sources and optical detectors and their use in the optical communication system.

References:

1. <https://www.youtube.com/watch?v=VG Gnv-gdoj8>
2. <https://www.exploratorium.edu/snacks/modulated-led>

3. Gerd Keiser, "Optical Fiber communication" MC Graw -Hill international, 4th Edition, 2010.
4. John M. Senior, "Optical Fiber communication", Second Edition, Pearson Education, 2007.